



TWO-FRAME SURVEY TELEMETRY

Defect ID: e05109f9-06e8-49d7-91f5-0e8ef448772a Current Scan: 2026-09-15 12:00:00 UTC
Coordinates: 28.613928, 77.209021 Time Delta: Initial Scan (Base Frame)

Two-Frame Spatial-Temporal Comparative Metrics

CURRENT DEFECT AREA

0.4500 m²
Prev: None (Base Frame)

AREA DELTA %

+0.0%
State: New

MUNICIPAL PRIORITY

Rank 8 / 10
Severity: High

Civil Engineering Advisory (Gemini 1.5 Flash)

Root Cause Analysis

Severe asphalt fatigue and moisture seepage under heavy axle loading.

Recommended Remediation

Full-depth asphalt patching and edge drainage restoration.

Pavement Condition Index (PCI) Technical Appendix

Standardized ASTM D6433 road survey assessment methodology applied. Structural integrity reflects localized shear stress and cyclic asphalt thermal contraction. Municipal action schedule requires hot-pour rubberized mastic application within 14 operational days to mitigate base-course gravel dispersion and water ingress.

Quality Assurance & Municipal Compliance

Survey verified through automated multi-modal visual vector comparison and RTK-calibrated GPS waypoint logging. All reported surface deformations adhere to AASHTO Highway Maintenance Guidelines for localized distress mitigation.



ScanTrail Engineering Certification & Audit

Municipal Pavement Condition & Verification Record

AASHTO / ASTM Compliance & Maintenance Matrix

Parameter	Measured Value	Threshold / Tolerance
Defect Identifier	e05109f9-06e8-49d7-91f5-0e8ef448772a	Unique UUID-4 Spatial Key
Geodetic Coordinate (Lat)	28.613928°	WGS-84 RTK < 0.05m tolerance
Geodetic Coordinate (Lon)	77.209021°	WGS-84 RTK < 0.05m tolerance
Surface Area Delta	+0.00%	Evolution Threshold: ±15.0%
Current Bounding Geometry	0.4500 m²	Calibrated Orthomosaic Grid
Previous Bounding Geometry	N/A	Baseline Historical Scan
Temporal Audit Interval	Initial Scan (Base Frame)	Dynamic Rover Visit Cycle
Municipal Priority Scale	8 / 10	1 (Restored) to 10 (Critical)
Target Work-Order Completion	14 Calendar Days	AASHTO Standard Maintenance Cycle

Municipal Engineer Sign-off & Verification Statement

This document certifies that the automated road survey conducted at the specified geodetic coordinates has been logged into the municipal spatial asset register. The multi-modal matching engine confirmed the temporal defect progression via two-frame vector alignment. The actionable recommendations provided are approved for procurement and contractor dispatch under standard municipal infrastructure maintenance protocol.

Authorized Municipal Road Inspector

ScanTrail Autonomous Telemetry Engine